

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Examination (CL/ME/EE/CE/IT/EC)

December 2012

CL 102 Environmental Sciences

Date: 13.12.2012, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) What is the meaning of environment? Explain the importance of man environment relationship. [03]
- (b) Explain the impact of technology and development on environment. [04]
- Q - 2 (a) Explain structure of atmosphere with neat sketch in detail. [04]
- (b) What are causes of Environmental Pollution? List all Environmental Pollution and their sources in detail. [05]
- (c) Explain the effects of air pollutants on human being and plant. [05]

OR

- Q - 2 (a) Enlist Component parts of Environment and explain in detail Hydrosphere. [04]
- (b) Discuss adverse effects of water pollution. [05]
- (c) Define noise. Briefly describe the sources and effects of noise pollution. [05]
- Q - 3 (a) Enlist the common air pollutants stating their permissible limits as per ambient air Quality standards. [04]
- (b) Enlist Green House Gases and explain 'Green House Effect'. [05]
- (c) Explain Eutrophication. Highlight causes and effects of Eutrophication on lake. [05]

OR

- Q - 3 (a) Give classification of water pollutants and describe organic water pollutants. [04]
- (b) What is 'Acid Rain'? Write a brief note on effects of acid rain on environment. [05]
- (c) Explain the term "environmental degradation" and discuss the role of advanced technology in the degradation of environment. [05]

SECTION - II

- Q - 4 (a) Explain the following terms. [03]
- a) Ecology. b) Nitrogen fixation. c) Denitrification.
- (b) Enlist components of an ecosystem. What is the role of consumers in an ecosystem? [04]
- Q - 5 (a) Explain in detail 'food web' with example. [04]

- (b) Define "Bio-Geo-Chemical" cycle. Give environmental significance of hydrological cycle. [05]
- (c) Explain causes and effects of deforestation. [05]

OR

- Q - 5 (a) Explain Energy flow with a neat sketch. [04]
- (b) What do you mean by Ecological Pyramids? Explain in detail "Pyramid of Number". [05]
- (c) Give advantages and disadvantages of solar energy. [05]
- Q - 6 (a) Define human settlement and enlist environmental factors affecting it. [04]
- (b) Which are the important theories of population growth? Explain Malthusian theory. [05]
- (c) Differentiate between Conventional and Non Conventional Energy Resources. [05]

OR

- Q - 6 (a) Explain Demographic Transition Theory. [04]
- (b) Enlist methods for population forecasting and explain how overpopulation is acting as pollutants. [05]
- (c) Explain problem due to overexploitation of water resources. [05]
